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Applications of higher order Markov models and Pressure Index to strategize controlled run chases in Twenty20 cricket

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Abstract

In limited overs cricket, the team batting first posts a target score for the team batting second to achieve in order to win the match. The team batting second is constrained by decreasing resources in terms of number of balls left and number of wickets in hand in the process of reaching the target as the second innings progresses. The Pressure Index, a measure created by researchers in the past, serves as a tool for quantifying the level of pressure that a team batting second encounters in limited overs cricket. Through a ball-by-ball analysis of the second innings, it reveals how effectively the team batting second in a limited-over game proceeds towards their target. This research employs higher order Markov chains to examine the strategies employed by successful teams during run chases in Twenty20 matches. By studying the trends in successful run chases spanning over 16 years and utilizing a significant dataset of 6537 Twenty20 matches, specific strategies are identified. Consequently, an efficient approach to successful run chases in Twenty20 cricket is formulated, effectively limiting the Pressure Index to [0.5, 3.5] or even further down under 0.5 as early as possible. The innovative methodology adopted in this research offers valuable insights for cricket teams looking to enhance their performance in run chases.

Keywords

Batting, home ground advantage, sport analytics, team strategy

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Introduction

Cricket is a team sport played between two teams consisting of 11 players each, following a set of rules as per the format of the game being Tests, One Day Internationals or Twenty20 Internationals (First Class, List A and Twenty20 respectively on a larger basis). In addition, shorter formats of the game like T10, The Hundred, etc., are being introduced for rapid spread and popularity of the game worldwide. One of the major reasons for the intense popularity of cricket nowadays is the uncertainty during a game and it has been observed that the shorter the format, the higher the uncertainty. The above claim is consistent with the fact that in the 5-year period from July 2014 to June 2019, 29.89% Test matches have been won by teams outside the top 3 ICC men's Test rankings against teams ranked within the top 3 ICC men's Test rankings, which increases to 33.67% if we consider Twenty20 cricket during the same period. The rise is even more in the following 5-year period from July 2019 to June 2024 where it increases from 19.48% in Tests to 29.30% in Twenty20 cricket.[1]

In Twenty20 cricket, a team bats for a maximum of 20 overs with 10 wickets in hand and tries to score as many runs as possible within the provided resources. The team which fields first then comes in to bat with identical resources, being set a target to score more runs than the team which batted first. Unless in rain-interrupted matches or other scenarios resulting in reduction of overs, the team which scores more runs wins the match. The match is called a tie in case both teams end up scoring equal number of

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runs. Super overs (earlier, bowl outs) are often played to decide the outcome of tied matches.

As the game progresses, efficient use of numbers, statistics and technology plays a key role to analyze and strategize the game from a player and team's point of view. The present study is an endeavor to introduce a strategy whereby teams could constantly set a target, within themselves, for their batters at certain intervals throughout the innings while chasing a score in a regular Twenty20 match with the help of the Pressure Index and higher order Markov chains. The Pressure Index is a ball-by-ball measure which quantifies the pressure experienced by the team batting second. It is a function of the required run rate and wickets lost at any point in the second innings, for our purposes, in a Twenty20 match. Thus, during a run chase, the value of the Pressure Index might change after every ball. The value of the Pressure Index starts at unity at the start of a run chase and keeps on changing as the innings progresses. It eventually reaches zero if the target is achieved but rises steeply if the team batting second fails to achieve the target. Details on working formulae, functionality, etc., regarding the Pressure Index are discussed later. The curve obtained by plotting the Pressure Index values against each ball during an innings is said to be the Pressure Curve. The Pressure Curve gives a clear indication of the status of the batting team at any given instance during a run chase. In this paper, Pressure Index values have been obtained corresponding to numerous Twenty20 matches spanned over a prolonged period. Then the concepts of Markov chains have been used in a way to learn the pattern of changes in the Pressure Index throughout a run chase in Twenty20 cricket and an attempt has been made to suggest a suitable strategy to keep the Pressure Index under control so that teams do not often lose grip on run chases.

Basic terminologies

A few basic terms prevalant in Twenty20 cricket are described as follows.

i.*Overs*: In cricket, the batsman stands at one end of the pitch, and the bowler delivers the ball from the other end. Now, after the bowler has delivered six consecutive legal balls, he is said to have completed an over.
ii.*Powerplay*: The first 6 overs of a Twenty20 match are called powerplay, and a maximum of two fielders can be placed outside the 30-yard circle during this period (see Talukdar[1]).
iii.*Death Overs*: The last few overs of any innings of a cricket match are said to be Death Overs. We will typically denote the last 4 overs of a Twenty20 innings to be Death Overs throughout the paper.
iv.*Middle Overs*: The overs following the powerplay till the beginning of the death overs is said to be Middle Overs.
v.*Openers*: Since batting takes place in pairs in cricket, the pair of batters who come in to bat at the start of an innings are termed as *Opening partners* or simply, openers (see Talukdar[1]).
vi.*Batting Depth*: If the bowlers of a team, who generally come to bat lower down the order, also possess good batting skills, then the team is said to have significant batting depth.
vii.*Tailenders*: Players who are participating in the team primarily for bowling skills and aren't a master in the art of batting are called Tailenders or *late-order batters* (see Knight and Dunmore[2]).
viii.*Target*: In limited overs cricket, the team batting first (refer as Team *A*) tries to score as many runs as they can in the allocated number of overs, without losing all their wickets. At the end of the innings, one run more than the total number of runs scored by Team *A* is the target for the opposition (Team *B*, say). Team *B* has to reach the target runs within the alloted number of overs in order to win the match.

Following a literature review and discussions on the novelty of this work in Section "Review of literature", in Section "Concepts of Markov chain", we present the fundamental concepts of Markov chains and motivate the need for higher-order models in the context of run-chase dynamics. Section "Definitions and formulation" introduces the Pressure Index in detail, outlines its computation, and demonstrates how its over-wise evolution reflects match situations. In Section "Estimating the order of Markov chain", we estimate the optimal order of the proposed Markov model. In Section "Discretization of pressure index", we discuss discretization of Pressure Index suggesting suitable gridding. Sections "Phase-wise Markov models" to "Applying the Markov model on Pressure Index" describe the construction of transition matrices, distributional modeling of PI across phases of an innings, and contextual factors such as home/away conditions and tournament-specific behavior. Building on these results, Section "Strategy for controlled run chases using the Markov model" develops a prescriptive framework that provides situation-specific Pressure Index targets and strategic guidelines for teams aiming to control run chases in Twenty20 cricket.

Review of literature

Though the paper is heavily dependent on the Pressure Index for formulating a strategic approach towards successful run chases in the Twenty20 format of cricket, the work of Bhattacharjee and Lemmer[3] and their subsequent extended works, to name a few, Talukdar and Bhattacharjee[4] and Bhattacharjee et al.[5] are not the only contributions in this domain of tactic building for run chases.

Some earlier works that were concerned with the progress of scoring in the second innings of limited overs

matches are Bailey and Clarke[6] and De Silva et al.[7] While Bailey and Clarke[6] attempted to predict the final score of the second innings of a One-day match while the game was in progress,[7] attempted to estimate the magnitude of victory in One-day matches. In both cases the authors used the Duckworth–Lewis resource table at the end of each over of the second innings of the match.

Choudhury and Bhargava[8] used Multilayer Perceptron Neural Networks to predict the outcome of a cricket tournament involving more than two teams. Sankaranarayanan[9] developed a prediction system that takes in historical ODI cricket match data as well as the instantaneous state of a match and predicts the score at key points in the future, culminating in a prediction of victory or loss. Using factors of the current situation of a cricket match,[10] attempted to predict the outcome through a two-stage process. In the first stage, the match situation and other cricketing attributes are used to predict the first innings score and then in the second stage the possibility of the team batting second achieving the target is estimated. Asif and McHale[11] presented a model for forecasting the outcomes of One-Day International cricket matches when the game is in progress using dynamic logistic regression. O'Riley and Ovens[12] compared four different score prediction tools by extending them to a ball-by-ball prediction model for predicting the final score, based on the present state of the first innings of a match. A series of papers are available in the literature that used different machine learning tools for predicting either the target of the match or the winner of the match when the match is in progress, viz.[13–16] Another set of researchers worked on the development of the Win and Score Predictor (WASP). The purpose of WASP is to predict the score at the end of the first innings of a limited overs match given the current situation and to predict the probability of a successful run chase of the team batting second given the target. Some literature concerning this includes.[17–19]

Most of the available literature regarding cricket match situation analysis are directed towards predicting final scores, match outcomes, or win probabilities using traditional or recent statistical or data mining tools. The models applied and comprehensive methodologies developed by previous researchers are highly valuable for forecasting. But such models are either descriptive or predictive in nature. However, in this current study, the researchers tried to develop a prescriptive model. By combining a measure of quantifying match pressure with a complex third-order Markov model, the current research attempts to frame strategies for successful run chases in Twenty20 cricket. The approach provides an exclusive, real-time tool for coaches to manage the pressure during run-chase on an over-by-over basis, which differentiates the contribution of the paper from the existing body of literature.

Objectives

The paper achieves the following objectives:

1. To develop a Markov model to quantify whether teams have exhibited any set pattern in successful run chases over a prolonged period of time.
2. To study the patterns of run chases and accordingly prepare strategies in order to keep teams batting second in control throughout run chases for future Twenty20 matches.

Concepts of Markov chain

Markov chain

Definition 1 A random process $\{X_n : n \geq 0\}$ is said to be a Markov chain with state space S if it satisfies the Markov condition, i.e., for any states $x_0, x_1, x_2, \ldots, x_{n-1}, i, j \in S$,

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = x_{n-1}, \ldots, X_1 = x_1, X_0 = x_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i) \quad (1)$$

It can be easily verified that the Markov condition is equivalent to the condition: for any states $x_1, x_2, \ldots, x_{k-1}, i, j \in S$,

$$\mathbb{P}(X_{n+1} = j \mid X_{n_k} = i, X_{n_{k-1}} = x_{k-1}, \ldots, X_{n_1} = x_1) = \mathbb{P}(X_{n+1} = j \mid X_{n_k} = i) \quad (2)$$

for any $n_1 < n_2 < \cdots < n_k \leq n$.

The evolution of the chain clearly depends on the probabilities $\mathbb{P}(X_{n+1} = j \mid X_{n_k} = i)$. In general, this depends on i, j and n. We restrict our attention to the case where this probability is independent of n.

Definition 2 A transition matrix $\mathbf{P} = ((p_{ij}))$ is the matrix whose (i, j)th entry is given by

$$((p_{ij})) = \mathbb{P}(X_1 = j \mid X_0 = i). \quad (3)$$

It is clear that, being conditional probabilities, the p_{ij} 's must satisfy

$$p_{ij} \geq 0 \quad \text{and} \quad \sum_{j \in S} p_{ij} = 1 \quad \text{for all} \quad i \in S. \quad (4)$$

Markov chains of higher order

The setup mentioned in Section "Markov chain", where the probability of an event at any time point depends only on that of the immediate past, can be thought of as a first order Markov chain. We may encounter situations where the probability of an event at a certain time point (say, the current state) might depend not just on the immediate past but on several time points prior to the current state. These are considered to be higher order Markov chains.

Table 1. Wicket weights assigned by Lemmer.

Position (i)	1	2	3	4	5	6	7	8	9	10	11
Wicket weight (w_i)	1.30	1.35	1.40	1.45	1.38	1.18	0.98	0.79	0.59	0.39	0.19

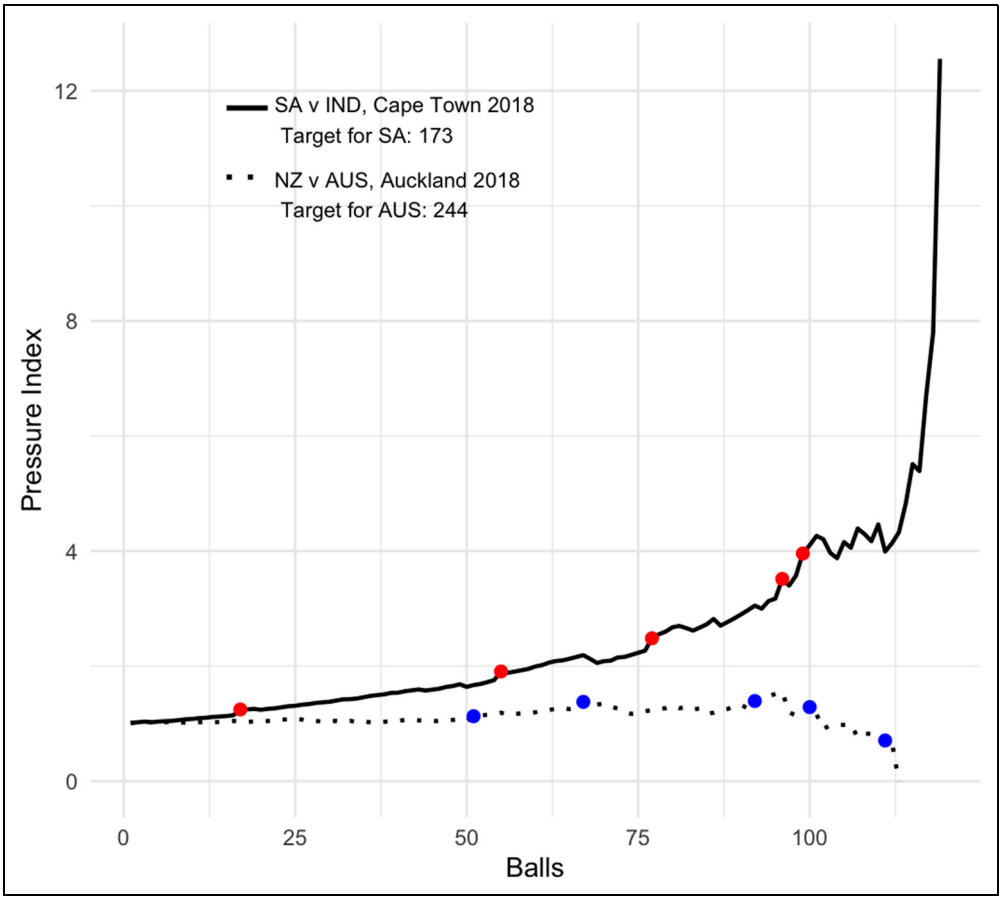

Figure 1. Pressure Curves for T20I 649 (NZ v AUS 2018) and T20I 655 (SA v IND 2018).
(Note: The dots in the pressure curves denote fall of wickets)

Definition 3 A random process $\{X_n : n \geq 0\}$ is said to be a Markov chain of order k if for any states $x_0, x_1, x_2, \ldots, x_{n-1}, i, j \in S$,

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, \ldots, X_0 = x_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i, \ldots, X_{n-k+1} = x_{n-k+1}). \quad (5)$$

For our purpose, consider $\{X_n : n \geq 1\}$ to be the sequence of Pressure Index values (see details in Section "Introduction to Pressure Index") for a team batting second at the end of each over during a run chase in a Twenty20 match, i.e., X_n is the Pressure Index value at the end of the $(n+1)$th over during a run chase in a Twenty20 match. We will see why such a sequence is a suitable choice for Markov modeling.

Introduction to Pressure Index

Definitions and formulation

As mentioned earlier, the Pressure Index (PI) is a ball-by-ball measure which quantifies the pressure experienced by the team batting second. In Twenty20 cricket, the team batting second has to achieve a certain target before its resources are exhausted. Thus, the pressure of the team batting second reduces if runs are scored at a

Table 2. Comaprison between models of different orders.

Order	Observations	Parameters (m_k)	Log likelihood	AIC	BIC	P-to-O ratio
1	116,383	33,034	−439.8439	945,755.9	1,265,017.7	0.2838
2	115,028	91,573	−372.0692	604,431.9	1,444,466.3	0.7961
3	110,025	101,829	−2.7725	197,590.4	950,321.2	0.9255
4	102,127	103,732	−6.8451	190,754.9	913,820.6	1.0157
5	90,309	90,321	−5.2042	189,364.6	870,646.3	1.00001
6	83,826	83,937	−1.3862	187,659.5	870,115.9	1.001
7	77,348	77,357	−1.0332	164,698.8	86,747.6	1.00001

Table 3. Expected outcomes of *PI* values based on previous states.

Pressure Index	Probability of occurance	Pressure Index	Probability of occurance	Pressure Index	Probability of occurance
0.00	0.010310341	0.01	0.020620683	0.02	0.005155171
0.31	0.010310341	0.66	0.010310341	0.82	0.010310341
0.88	0.010310341	1.21	0.010310341	1.25	0.010310341
1.36	0.010310341	1.39	0.010310341	1.40	0.010310341
1.42	0.005155171	1.43	0.010310341	1.46	0.010310341
1.47	0.010310341	1.48	0.010310341	1.51	0.015465512
1.52	0.015465512	1.54	0.015465512	1.56	0.005155171
1.57	0.010310341	1.58	0.015465512	1.59	0.036086194
1.60	0.018867925	1.61	0.020620683	1.62	0.020620683
1.64	0.034333436	1.66	0.041241365	1.67	0.030931024
1.68	0.041241365	1.69	0.020620683	1.70	0.015465512
1.71	0.030931024	1.72	0.046396536	1.73	0.046396536
1.74	0.025775853	1.75	0.030931024	1.76	0.056706877
1.79	0.005155171	1.80	0.010310341	1.81	0.030931024
1.82	0.020620683	1.84	0.030931024	1.86	0.041241365
1.87	0.010310341	1.90	0.010310341	1.91	0.010310341
1.94	0.003402413	1.95	0.005155171	2.00	0.010310341
2.02	0.020620683	2.23	0.010310341	2.39	0.010310341

fast rate and if a considerable number of wickets are kept in hand until the target is reached. Based on this idea, the formula of the Pressure Index[3] for teams batting second in Twenty20 matches is given by:

$$PI = \frac{CRRR}{IRRR} \times \frac{1}{2}\left[e^{\frac{RU}{100}} + e^{\frac{\sum w_i}{11}}\right] \tag{6}$$

where *IRRR* is the initial required run rate, i.e., if T is the target to be scored by the team batting second in B balls, then

$$IRRR = \frac{T \times 6}{B}. \tag{7}$$

CRRR denotes the current required run rate at any point during the second innings. If R' runs have been scored in B' balls at some stage, then

$$CRRR = \frac{(T - R') \times 6}{B - B'}. \tag{8}$$

The ratio $\frac{CRRR}{IRRR}$ thus measures the progress of the team batting second during a run chase. A lower value of the ratio denotes that the team is supposedly ahead in the chase.

The term $\sum w_i$ denotes the sum of the weights of the wickets that have fallen until any stage of the innings. As wickets fall, the team's wicket strength deteriorates. For example, if a top order batter gets out, the wicket strength of the team drops more than when a lower order batter gets out. In order to account for the varying abilities of batters affecting their wicket strengths, wicket weights (w_i) as per[20] are used instead of assigning equal weights to every wicket. The wicket weights for different batting positions, as assigned by Lemmer,[20] are given Table 1.

Finally, *RU* denotes the percentage of resources used by the team at any stage during the innings as per the Duckworth–Lewis table.

Several other formulae to calculate the Pressure Index are available but we will avoid using those in this study due to various limitations. One such formula, as suggested by Shah and Shah,[21] is as follows:

$$PI = CI \times 100 + \left[\frac{wicket\,weight}{180} \times T \times \frac{Br}{B} \times \frac{Rr}{T}\right] \tag{9}$$

Table 4. Few common transitions in order-3 model with 0.01 precision.

PI_{t-3}	PI_{t-2}	PI_{t-1}	PI_t	No. of occurances
1.02	1.04	1.10	1.15	4
1.03	1.05	1.07	1.11	4
0.97	0.98	0.95	1.01	3
0.97	1.03	1.01	1.01	3
0.98	0.99	0.96	0.97	3
0.98	1.05	1.12	1.15	3
0.99	0.94	0.98	1.07	3
0.99	1.01	1.02	1.09	3
0.99	1.03	1.10	1.16	3

Table 5. Comparison between models with various discretization levels.

Precision	$N_{transitions}$	Singleton %	MAE	RMSE	Coverage %
0.01	81,519	98.86	1.155	4.344	30.854
0.025	68,576	86.02	0.904	4.308	78.679
0.05	42,445	79.31	0.878	4.296	84.404
0.075	31,198	58.64	0.758	4.263	93.244
0.1	22,884	37.44	0.671	4.096	97.248
0.15	18,465	31.20	0.672	4.122	97.692
0.2	11,181	29.88	0.672	4.131	97.931
0.25	10,336	28.44	0.674	4.131	98.048
0.5	3,458	16.36	0.673	4.135	99.414

According to the above formula, the PI value will always be 100 at the start. After calculating PI values for a considerable number of matches using this formula, it was found that PI decreased when a wicket fell and when the number of runs scored in an over was less than the required run rate. This is completely unrealistic considering the very purpose of the formulation of the Pressure Index and hence we will not take into account this formula.

Two other formulae which were also suggested, but had significant limitations, are given below.

$$PI_1 = \frac{CRRR}{IRRR} \times e^{\frac{\sum w_i}{11}} \tag{10}$$

$$PI_2 = \frac{CRRR}{IRRR} \times e^{\frac{RU}{100}} \tag{11}$$

The factor $CI = \frac{CRRR}{IRRR}$ is expected to change after every ball. While $e^{\frac{\sum w_i}{11}}$ changes only if a wicket falls, $e^{\frac{RU}{100}}$ changes after every ball as RU takes into consideration not only the number of wickets fallen but also the balls faced by a team. At the start of the second innings, the Pressure Index of a team is equal to 1 according to both PI_1 and PI_2 and it will fluctuate as the match progresses. If the team batting second is able to achieve the target, both PI_1 and PI_2 become 0. However, it seems that PI_2 is more sensitive compared to PI_1 because it depends on RU, which is expected to change after every ball throughout the innings.

Considering all these factors, the formula (6) seems to be the most commendable, and we will consider the same for further study in this paper.

Operational mechanism

It is evident from (6) that the Pressure Index value of a team batting second remains at unity at the start of a run chase. With every delivery bowled, the value might change and it gradually reaches zero once the team has achieved its target. In case the team fails to do so, the PI value rises steeply at the later stages of the match. To better understand how the indices work, the example of two matches have been taken.

Consider the Twenty20 match played between South Africa and India at Newlands Cricket Ground, Cape Town on February 24, 2018. South Africa started slowly but cautiously, chasing 173, 45/1 after 9 overs. Although the required run rate was creeping up, 9 wickets were still in hand and they were still in the game. But just when they started to accelerate, wickets began falling, which resulted in a halt in their scoring rate leading to a rapid increase in the Pressure Index. By the 18th over, SA completely lost their grip on the run chase and finally fell short by 7 runs even after scoring 45 runs off their last 3 overs. On the other hand, Australia successfully chased down a record 244 runs, just a week prior to that, at Auckland with the strategy of never letting the required run rate rise too high (Figure 1).

The urgency to score runs during a run chase increases as the game progresses, unless the target is too small to create any trouble for the team batting second. Also, scoring rates might halt when a wicket falls. The number of runs scored and wickets lost until any stage of the innings influence the scoring rate of the team in the next over. Furthermore, the target, the runs scored, and wickets lost until any stage of the innings determine the Pressure Index of the team at that stage. Thus, the Pressure Index at the end of an over during any stage of the innings depends on the Pressure Index of the team in the previous overs.

Thus, we can consider a sequence $\{X_n : n \geq 1\}$ where X_n is the Pressure Index value at the end of the $(n+1)$th over during a run chase in a Twenty20 match. This leads to a setup where the concepts of Markov chains on the sequence $\{X_n : n \geq 1\}$ can be purposefully implemented.

Methodology

Estimating the order of Markov chain

We have defined our sequence to be the chronological PI values at the end of each over of the second innings of a Twenty20 match but the number of preceding PI values on which the PI at a certain stage of the sequence depends is still unknown. In an ideal scenario, the runs scored and

Table 6. Phase-wise marginal distributional analysis of Pressure Index.

Phase	Mean *PI*	Median *PI*	SD_{PI}	25th percentile	75th percentile	IQR
Powerplay	1.323	1.301	0.218	1.17	1.46	0.29
Middle Overs	1.74	1.692	0.412	1.45	1.99	0.54
Death Overs	2.859	2.641	1.488	1.77	3.68	1.91

wickets lost in a certain over should technically depend on that in each of the previous overs in the innings. However, in order to avoid unnecessarily complex calculations and to achieve an optimum level of accuracy, one should try to find out the desired order of the Markov chain, i.e. the desired number of preceding *PI* values on which the next *PI* value in a sequence truly depends, that fits our data well and leads to a high enough precision for prediction purposes.

All men's Twenty20 matches in which the team batting second have either reached the target after batting for more than 18 overs or failed to reach the target and have lost the match by a margin not more than 10 runs have been considered for the study. Since the goal of our study is to prepare a strategy in support of teams batting second in Twenty20 matches, a certain section of the selected sample of matches will be helpful in replicating the patterns of successful run chases. The choice of such selection has also been made based on the idea of using a significantly large training sample in order to enhance the construction of the model.

Starting from the first ever officially recorded men's Twenty20 match on June 13, 2003 till October 15, 2025, a total of 8241 matches have been played where the outcomes have been either of the ones mentioned earlier. After removing matches in which complete ball-by-ball data are unavailable, we finally take into account a filtered set of 6537 matches[2] to create a set of 6537 sequences of the form $\{s_n : n \geq 1\}$. The correct order of the Markov chain is then estimated using these sequences to proceed with further study.

For an order-k Markov chain, the state space (S) has been defined to be the collection of all possible observed *PI* values. Based on the probabilities of transitions,

$$(s_{t-k}, s_{t-(k-1)}, \ldots, s_{t-1}) \rightarrow s_t$$

transition matrices have been constructed. For a match with T overs, an order-k Markov chain requires $(T-k)$ observations, and so each match contributes $(T-k)$ transitions. In our dataset, an average of 19.1 overs have been played per match. For example, for an order-3 Markov chain, we have approximately 104592 transitions[3] in the pooled data. Note that each match is treated independently, so transitions do not cross match boundaries.

Negative *PI* values are not of interest since they do not provide any inferential interpretation, so keeping them in analysis distorts distributional fits and transition matrices. Common statistical methods to handle them include censoring, truncation/flooring or redefining the state space etc. A simple alternative would be to set $PI = 0$ immediately as the target in a run chase is achieved. This essentially results in censoring where we proceed our analysis with $PI^+ = max(PI, 0)$, instead of raw *PI* which can take negative values. For brevity, we will denote PI^+ to be simply *PI* throughout the study. It should be ensured that a previous state ending with $s_{t-1} = 0$ cannot predict a non negative *PI* value (see details in Section "Discretization of pressure index").

Table 7. Results for Pressure Index in powerplay.

Criteria	Gamma	Exponential	Weibull
AIC	−1267.731	6438.255	−718.434
BIC	−1264.717	7119.283	−724.320
K-S test statistic (*D*)	0.026477	0.48476	0.055296

Table 8. Results for Pressure Index in middle overs.

Criteria	Gamma	Exponential	Weibull
AIC	632.815	6192.033	965.436
BIC	653.670	6186.024	971.667
K-S test statistic (*D*)	0.013766	0.400127	0.047228

Table 9. Results for Pressure Index in death overs.

Criteria	Gamma	Exponential	Weibull
AIC	1977.545	7019.481	3118.065
BIC	1992.033	7439.283	3126.974
K-S test statistic (*D*)	0.017849	0.32761	0.06927

Table 10. Estimated parameters of Gamma distribution.

Phases	Estimated shape parameter	Estimated rate parameter
Powerplay	38.276	28.931
Middle Overs	18.447	10.62
Death Overs	3.667	1.286

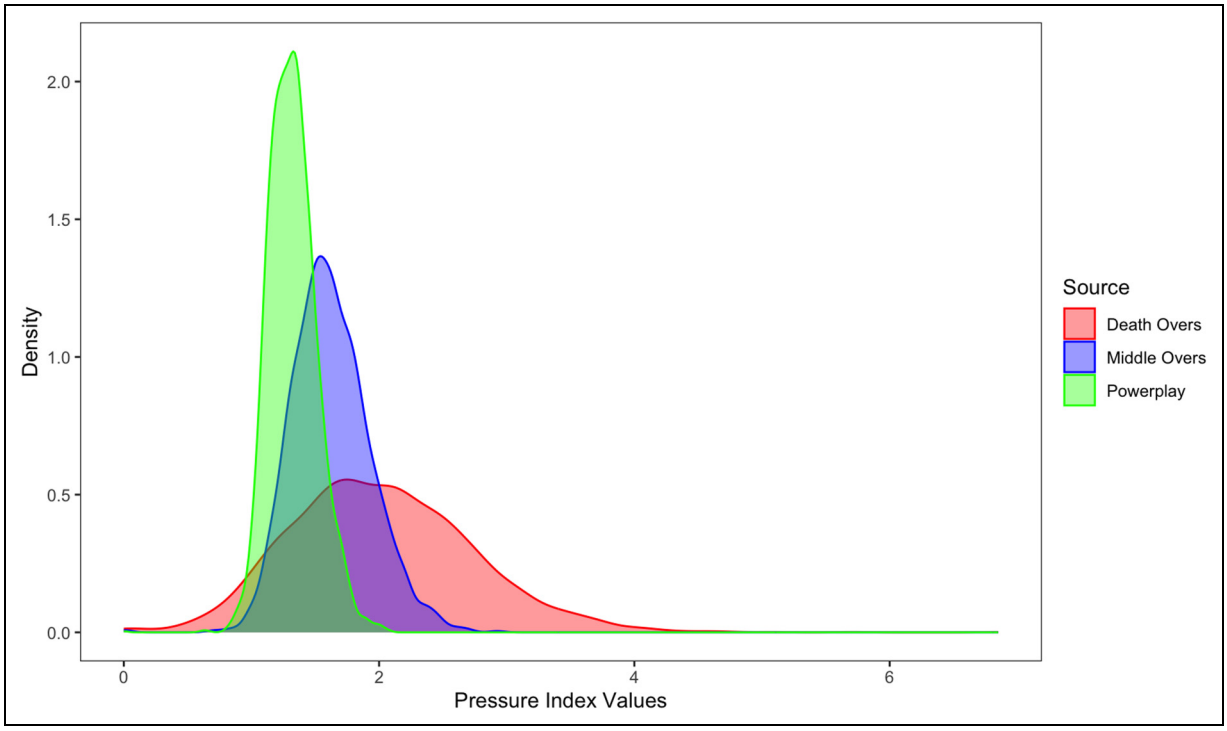

Figure 2. Density plots of phase-wise Pressure Index.

For a total of $M(=6537)$ matches in our dataset,

$$N_{transitions} = \sum_{m=1}^{M} (T_m - k) \tag{12}$$

For an order-k Markov chain, construct empirical probabilities

$$\mathbb{P}(s_t | s_{t-k}, s_{t-k+1}, \ldots, s_{t-1}) = \frac{N(s_{t-k}, s_{t-k+1}, \ldots, s_{t-1}, s_t)}{N(s_{t-k}, s_{t-(k-1)}, \ldots, s_{t-1})}, \tag{13}$$

where N denotes the count of transitions in the training data. Log likelihood function of the Markov model is given by,

$$L_k = \sum_{i=1}^{N_{transitions}} \mathbb{P}(s_t^{(i)} | \text{previousstate}^{(i)}) \tag{14}$$

Common model selection criteria include *AIC* and *BIC* which are defined as follows.

Table 11. Comparison of global vs phase-wise markov models.

	MAE		RMSE	
Phase	Global	Phase-wise	Global	Phase-wise
Powerplay	0.944	0.912	6.23	5.74
Middle Overs	0.877	0.209	3.03	0.292
Death Overs	0.173	0.151	0.316	0.197

$$AIC_k = 2m_k - 2L_k \tag{15}$$

$$BIC_k = m_k log(N_{transitions}) - 2L_k \tag{16}$$

where m_k is the number of free parameters, i.e. the number of unique transitions in the transition matrix of the corresponding model.

The entire data was treated as an 80%–20% split of training data and test data respectively and the results, from Table 2, show an order-3 model to have the highest log likelihood. Only models of order 3 or lesser have a parameter-to-observations ratio lesser than 1, resulting in significant overfitting for higher order models. Although *AIC* and *BIC* keep on decreasing as order increases, they already approach a certain saturation point for an order-3 model. The extreme overfitting restricts the predictive accuracy for higher order models as well. As a result of a trade-off between the factors, we consider an order-3 model to proceed for further study. A pseudo-code for obtaining the optimal order of a Markov chain is given in Algorithm 2.

An average batting pair (a partnership), in our dataset, lasts for 18.0126 (≈18) balls during a Twenty20 innings. For every break in partnership, a wicket falls, so *RU* and $\sum w_i$ change leading to a drastic change in *PI*. Saikia et al.[22] explained how partnerships and *PI* are related in Twenty20 matches. So heuristically, an order-3 Markov chain is expected to fit our dataset well. On the other hand, based on data from July 01, 2015[4] to October 15, 2025, an average batting pair lasts for 35.249 (≈35) balls

Table 12. Win rates induced by phase-wise *PI* thresholds.

Phases	*PI* thresholds	Home team win % (*H*%)	Away team win % (*A*%)	$\Delta = H - A$
Powerplay	[0, 0.5)	100	100	0%
	[0.5, 1)	73.7	62.2	+11.6%
	[1, 1.5)	70.6	61.8	+8.8%
	[1.5, 2)	68.2	58.5	+9.7%
	[2, 2.5)	42.7	36.6	+6.1%
	[2.5, 3)	14.6	11.1	+3.5%
Middle Overs	[0, 0.5)	100	100	0%
	[0.5, 1)	100	98.4	+1.6%
	[1, 1.5)	75.7	70.7	+5.0%
	[1.5, 2)	51.3	43.8	+7.6%
	[2, 2.5)	28.8	22.1	+6.6%
	[2.5, 3)	12.4	12.1	+0.3%
	[3, 3.5)	5.9	2.1	+3.9%
	[3.5, 4)	1.2	1.5	−0.3%
	[4, 4.5)	1.2	0	+1.2%
	[4.5, 5)	0	0	0%
Death Overs	[0, 0.5)	100	100	0%
	[0.5, 1)	100	98.8	+1.2%
	[1, 1.5)	94.4	94.3	+0.1%
	[1.5, 2)	85.7	81.6	+4.1%
	[2, 2.5)	67.5	60.2	+7.3%
	[2.5, 3)	55.7	47.5	+8.2%
	[3, 3.5)	36.4	14.7	+21.7%
	[3.5, 4)	25.9	11.1	+14.8%
	[4, 4.5)	8.8	0.6	+8.2%
	[4.5, 5)	3.5	0.9	+2.6%

Table 13. Win rates induced by over-wise *PI* thresholds.

Over	*PI* thresholds	Home team win % (*H*%)	Away team win % (*A*%)	$\Delta = H - A$
End of 16th over	[0, 0.5)	100	100	0%
	[0.5, 1)	100	100	+0%
	[1, 1.5)	94.2	94.1	+0.1%
	[1.5, 2)	86.6	85.6	+1%
	[2, 2.5)	70.2	56.9	+13.2%
	[2.5, 3)	38.8	29.4	+9.4%
	[3, 3.5)	18.8	7.4	+11.4%
	[3.5, 4)	4.3	3.5	+0.8%
	[4, 4.5)	3	0	+3%
	[4.5, 5)	0	0	0%
End of 17th over	[0, 0.5)	100	100	0%
	[0.5, 1)	100	100	0%
	[1, 1.5)	94.7	91.3	+3.4%
	[1.5, 2)	88.6	87.2	+1.4%
	[2, 2.5)	70.9	62.7	+8.2%
	[2.5, 3)	56.9	45.7	+11.2%
	[3, 3.5)	29.3	6.9	+22.4%
	[3.5, 4)	11.4	4.1	+7.3%
	[4, 4.5)	5.6	0	+5.6%
	[4.5, 5)	2.5	0	+2.5%
End of 18th over	[0, 0.5)	100	100	0%
	[0.5, 1)	98.8	93.4	+5.4%
	[1, 1.5)	90.5	89.9	+0.6%
	[1.5, 2)	82.9	78.1	+4.8%
	[2, 2.5)	76.6	66.7	+9.9%
	[2.5, 3)	60.7	51.6	+9.1%
	[3, 3.5)	34.8	20	+14.8%
	[3.5, 4)	27.3	14.2	+13.1%
	[4, 4.5)	10.5	0.6	+9.9%
	[4.5, 5)	3.6	0.9	+2.7%
End of 19th over	[0, 0.5)	100	100	0%
	[0.5, 1)	100	91.8	+8.2%
	[1, 1.5)	89.4	87.3	+2.1%
	[1.5, 2)	73.3	66.8	+6.5%
	[2, 2.5)	60.1	53.2	+6.9%
	[2.5, 3)	61.5	47.5	+14%
	[3, 3.5)	41.2	28.6	+12.6%
	[3.5, 4)	25	22.2	+2.8%
	[4, 4.5)	8.8	0.6	+8.2%
	[4.5, 5)	3.7	0.9	+2.8%

in a men's One Day International innings, in matches where a team batting second have either reached the target after batting for more than 45 overs or failed to reach the target and have lost the match by a margin not more than 15 runs. Moreover, a different Duckworth-Lewis table is used to extract *RU* and different Wicket weights have been defined by Lemmer,[23] for One Day International (or in general 50-over) matches. So, an order-6, instead of an order-3 Markov model, is expected to yield meaningful results for 50-over formats.

Discretization of Pressure Index

After the transition matrix for an order-3 model has been obtained, any new sequence consisting of k entries ($3 \leq k \leq T_j$), for some j-th match, can be inserted to predict the outcome of the unknown $(k+1)$th entry of the sequence. While a tuple of previous states is inserted, the model searches for occurrences of the exact set of previous states in the transition matrix. If multiple current states are observed corresponding to a tuple of previous states with non zero probabilities in the transition matrix, then the expected value of those observations is considered as the predicted outcome. However, if no occurrences of current states are found with non zero probability for a tuple of previous states, then a robust method to estimate the next *PI*, as described later in Sections "Phase-wise Markov models" and "Applying the Markov model on Pressure Index", would be to consider confidence intervals of *PI* values and look to estimate a Pressure Index within a certain range with high confidence, using a distributional (here, gamma) fallback.

Equation 6 treats *PI* values as continuous entities, hence taking values in an uncountable set. However in practice, an implicit discretization of *PI* being rounded to two decimal places, i.e. a 0.01 precision (or grid) has been performed.

The following example shows the outcome when (1.49, 1.63, 1.48) is entered as the new sequence. In

Table 14. Tournament-wise performance of the Markov model.

Competition	MAE	RMSE	Coverage %	Markov Usage %
IPL	0.8885	7.8697	84.16	95.86
T20 Blast	6.4472	14.1590	83.37	95.90
BPL	1.0662	4.5498	80.58	96.12
CPL	5.5460	8.9030	84.31	95.10
PSL	0.6081	3.6784	84.53	95.93
SMAT	1.6174	3.3080	84.97	95.66
SA20 League	6.3580	13.7701	84.16	93.70

BPL: Bangladesh Premier League; SMAT: Syed Mushtaq Ali Trophy.

cricketing terms, this translates to a situation where the Pressure Index of a team batting second have been recorded as 1.49, 1.63 and 1.48 at the end of three consecutive overs during the second innings of a Twenty20 match. Given the target and the number of overs the innings is into, we are interested in predicting the probable Pressure Index of the team at the end of the next over (see pseudo code in Algorithm 4). Table 3 provides the list of all possible *PI* values a team has exhibited in the training sample after having a previous state *PI* tuple (1.49, 1.63, 1.48) along with the corresponding probabilities and the resulting expected *PI*.

Table 3 tells us that if a team faces a situation where they have recorded Pressure Index values of 1.49, 1.63, 1.48 in that order at the end of three consecutive overs during the second innings of a Twenty20 match, then the expected Pressure Index value at the end of the next over would be approximately 1.59.

Inspite of a large number of transitions, finding exact replication of a 3-tuple of *PI* values from the transition matrix might be difficult and in some cases, impossible. Table 4 shows that no set of 4 consecutive *PI* values have occured more than 4 times in the matrix. Even after using a 0.01 grid, 98.86% of the unique transitions appear only once in the matrix, indicating severe sparsity. To address this, Laplace smoothing was implemented which did not yield satisfactory results due to high proportion of singleton occurances, extremely low proportion (0.11%) of reliable states, i.e. states with 10 or more observations, and only a 30.85% coverage on the test data.

As a robust alternative, several precisions ranging from 0.01 to 0.5 were tested (see Algorithm 3) and the results are shown in Table 5. Based on the least MAE and RMSE values and a trade-off between proportions of singleton occurances and coverage probabilities, which signifies coarseness of the data, a 0.1 precision has been chosen for further study.

Phase-wise Markov models

Although Sections "Estimating the order of Markov chain" and "Discretization of pressure index" provide insights into

Table 15. Predictions on IPL, BBL and PSL based on the Markov model for IPL data.

	IPL				BBL				PSL			
Phase	MAE	RMSE	Coverage	Markov usage	MAE	RMSE	Coverage	Markov usage	MAE	RMSE	Coverage	Markov usage
Powerplay	0.102	0.129	86.5	100	2.209	8.236	85.4	100	2.112	7.144	83.5	98.8
Middle Overs	0.165	0.219	93.7	100	1.178	0.293	89.4	99.8	2.18	0.363	92.4	99.7
Death Overs	0.296	0.856	83.2	98.6	0.313	1.913	78.5	97.5	0.342	1.78	77.5	94.8

Table 16. Predictions on IPL, T20is and CPL based on the markov model for BBL data.

	IPL				T20Is				CPL			
Phase	MAE	RMSE	Coverage	Markov usage	MAE	RMSE	Coverage	Markov usage	MAE	RMSE	Coverage	Markov usage
Powerplay	1.097	6.124	82.7	100	2.115	8.151	78.4	99.6	1.104	7.132	81.6	100
Middle Overs	1.183	1.307	92.5	99.5	0.247	0.396	89.5	98.3	1.205	0.355	90.6	99.3
Death Overs	0.895	0.856	70.1	92.1	1.355	1.996	63.1	91.2	1.914	0.699	79.9	82.7

how Pressure Index values can be monitored throughout an innings, it emerges as quite a general approach irrespective of a team's specific patterns during run chases in certain situations. For a better understanding, one might be interested in studying the patterns of where a team's Pressure Index lie at different stages during a successful run chase in Twenty20 cricket.

Table 6 shows significant behavioral differences in *PI* values during Powerplay (Overs 1 to 6), Middle Overs (Overs 7 to 16) and Death Overs (Overs 17 to 20) of innings during several successful run chases (comprising 4083 out of the 6537 matches from our initial training set). The choice of these specific phases are quite common in practice and are often found in literature for various phase-wise analysis of a Twenty20 innings. Variance of the marginal distribution of Pressure Index being the largest at Death Overs is in terms with the fact that uncertainity in a run chase generally increases as the innings progresses. Kolmogorov-Smirnov tests have been then performed over a few known distributions, and AIC and BIC have been observed (see Algorithm 5) to determine which known distribution fits the data corresponding to *PI* values the best (Tables 7 to 9).[5]

Lower AIC, BIC and Kolmogorov-Smirnov test statistic values denote that the gamma distribution fits well for the *PI* values corresponding to each section of the data. This gives us an idea of how the *PI* values at three different stages of an innings are distributed.[6]

Using the estimated gamma parameters in Table 10, the phase means are obtained and the variance increases markedly towards the death overs. Practically, this means *PI* is relatively concentrated near its phase mean during the Powerplay and becomes more dispersed as innings progresses. We exploit this in two ways within our strategy: (a) when empirical Markov transitions are sparse for a given tuple, the fitted gamma provides a phase-specific fallback predictive distribution for *PI* which is then used to compute expected PI and confidence intervals, and (b) the gamma quantiles are used to map *PI* ranges into the suitable zones (see details in Section "Strategy for controlled run chases using the Markov model"). Thus, besides descriptive overlays, the gamma fits provide phase-specific probabilistic approximations that increase coverage when transition data are missing and frame a actionable strategy by linking percentiles to tactical thresholds (Figure 2).

The Kolmogorov-Smirnov statistic D is reported only for descriptive comparison. Since parameters are estimated from the same data, classical K-S p-values are invalid. A valid goodness-of-fit assessment is therefore performed using a parametric bootstrap (see Algorithm 6), which refits the model to bootstrap samples to obtain the empirical null distribution of D.

Separate transition matrices corresponding to the three different phases of an innings can be constructed based on the process described in earlier sections and outcomes resulting from a tuple consisting of three *PI* values will then be

Table 17. Pressure Index of Pakistan: vs West Indies, April 03, 2018.

			0.01 precision			0.1 precision		
Overs	Cumulative runs	Cumulative wickets	Actual PI	Predicted PI	95% C.I	Actual PI	Predicted PI	95% C.I
1	5	0	1.04	–	–	1.0	–	–
2	15	0	1.04	–	–	1.0	–	–
3	32	0	0.99	–	–	1.0	–	–
4	51	0	0.91	1.09	[0.94, 1.24]	0.9	1.0	[0.9, 1.2]
5	57	0	0.93	0.82	[0.67, 0.97]	0.9	0.8	[0.7, 1.0]
6	62	1	1.04	0.82	[0.67, 0.97]	1.0	0.8	[0.7, 1.0]
7	71	1	1.03	0.92	[0.78, 1.06]	1.0	0.9	[0.7, 1.1]
8	79	1	1.03	0.94	[0.80, 1.08]	1.0	0.9	[0.7, 1.1]
9	86	1	1.05	1.05	[0.86, 1.24]	1.1	1.0	[0.9, 1.2]
10	89	1	1.13	1.13	[0.97, 1.29]	1.1	1.1	[1.0, 1.3]
11	98	1	1.11	1.08	[0.94, 1.22]	1.1	1.1	[1.0, 1.3]
12	107	1	1.08	1.09	[0.95, 1.23]	1.1	1.1	[1.0, 1.3]
13	114	2	1.15	1.14	[0.97, 1.31]	1.2	1.1	[1.0, 1.3]
14	120	2	1.17	1.18	[1.03, 1.32]	1.2	1.2	[1.1, 1.4]
15	132	2	0.83	0.83	[0.67, 0.99]	1.0	0.8	[0.7, 1.0]
16	139	2	0.62	0.63	[0.47, 0.79]	1.0	0.6	[0.5, 0.8]
17	154	2	0	0.06	[0.00, 0.27]	0	0	[0.0, 0.2]

Table 18. Pressure Index of Chennai Super Kings: vs Delhi Capitals, October 10, 2021.

			0.01 precision			0.1 precision		
Overs	Cumulative runs	Cumulative wickets	Actual PI	Predicted PI	95% C.I	Actual PI	Predicted PI	95% C.I
1	8	1	1.08	–	–	1.0	–	–
2	16	1	1.11	–	–	1.1	–	–
3	20	1	1.24	–	–	1.2	–	–
4	34	1	1.13	1.15	[1.01, 1.29]	1.1	1.2	[1.1, 1.4]
5	39	1	1.19	1.18	[1.04, 1.33]	1.2	1.2	[1.1, 1.4]
6	59	1	1.14	1.21	[0.99, 1.28]	1.1	1.2	[1.1, 1.4]
7	64	1	1.21	1.23	[1.00, 1.31]	1.2	1.2	[1.1, 1.4]
8	68	1	1.30	1.21	[0.98, 1.29]	1.3	1.2	[1.1, 1.4]
9	75	1	1.36	1.35	[1.19, 1.51]	1.4	1.4	[1.3, 1.6]
10	81	1	1.45	1.46	[1.31, 1.61]	1.5	1.5	[1.4, 1.7]
11	94	1	1.43	1.49	[1.35, 1.63]	1.4	1.5	[1.4, 1.7]
12	99	1	1.56	1.46	[1.31, 1.61]	1.6	1.5	[1.4, 1.7]
13	111	1	1.51	1.67	[1.52, 1.82]	1.5	1.7	[1.6, 1.9]
14	117	3	1.69	1.56	[1.40, 1.72]	1.7	1.6	[1.5, 1.8]
15	121	4	2.29	1.97	[0.83, 1.11]	2.3	2.0	[1.9, 2.2]
16	129	4	2.49	2.46	[2.31, 2.61]	2.5	2.5	[2.4, 2.7]
17	138	4	2.68	2.52	[2.34, 2.70]	2.7	2.5	[2.4, 2.7]
18	149	4	2.85	2.79	[2.62, 2.97]	2.9	2.8	[2.7, 3.0]
19	160	5	3.39	3.61	[3.42, 3.80]	3.4	3.6	[3.5, 3.8]
20	173	6	0	4.40	[4.21, 4.59]	0	4.4	[4.3, 4.6]

obtained based on the transition matrix corresponding to the specific section of the innings only. Results in Table 11 clearly imply that phase-wise models exhibit better performance than a single generalized global model in terms of lesser MAE and RMSE.

Models based on the three-phase partition of the innings is expected to avoid the overgeneralization of teams' strategies in run chases caused by the global model (see details in Section "Applying the Markov model on Pressure Index").

Contextual factors in Pressure Index dynamics

Discrepancies in Pressure Index for home and away teams. Besides the three-phase partitioning of a Twenty20 innings, several external factors like home ground advantage,

Table 19. Situation-specific Pressure Index target recommendations.

		Home teams		Away teams	
Phases	*PI* thresholds	Win rate	Recommendation	Win rate	Recommendation
Powerplay	[0, 0.5)	100%	Target Zone	100%	Target Zone
	[0.5, 1)	73.7%	Acceptable	62.2%	Acceptable
	[1, 1.5)	42.7%	Risky	36.6%	Avoid
	[1.5, 2.5)	13.9%	Avoid	10.5%	Avoid
	[2.5, ∞)	0%	Avoid	0%	Avoid
Middle Overs	[0, 0.5)	100%	Target Zone	100%	Target Zone
	[0.5, 1)	100%	Target Zone	98.4%	Target Zone
	[1, 1.5)	75.7%	Acceptable	70.7%	Acceptable
	[1.5, 2.5)	43.6%	Risky	35.9%	Avoid
	[2.5, ∞)	6.9%	Avoid	3.7%	Avoid
Death Overs	[0, 0.5)	100%	Target Zone	100%	Target Zone
	[0.5, 1)	100%	Target Zone	96.8%	Target Zone
	[1, 1.5)	87.3%	Acceptable	70.2%	Acceptable
	[1.5, 2.5)	75.9%	Acceptable	68.2%	Acceptable
	[2.5, ∞)	11.2%	Avoid	8.1%	Avoid

winning the toss ahead of a match, team strengthening due to player-player coalition or player-coach coalition etc. might play crucial roles in determining teams' strategy for a successful run chase in Twenty20 matches. In our dataset, around 12.52% of the matches have been played at neutral venues (e.g. South Africa vs India at Perth Stadium on October 30, 2022) or at venues which are home grounds for both the participating teams (e.g. Bangladesh vs Bangladesh A at Sher-E Bangla National Stadium, Mirpur on December 11, 2013). Excluding this section of matches, home teams have won 53.93% of the games, which rises up to 57.42% if we consider Twenty20 International matches only.

Table 20. Over-wise predictive performance of the Markov model.

Overs	Mean actual *PI*	Mean predicted *PI*	Coverage	MAE	RMSE
4	1.13	1.17	99.7%	1.162	6.189
5	1.21	1.24	99.2%	1.031	5.220
6	1.32	1.32	98.4%	0.470	5.244
7	1.41	1.45	98.1%	0.184	0.294
8	1.52	1.54	94.4%	0.190	0.349
9	1.63	1.68	92.6%	0.207	0.137
10	1.75	1.84	88.1%	0.233	0.186
11	1.88	1.94	82.0%	0.298	0.548
12	2.03	2.12	89.2%	0.502	0.690
13	2.19	2.32	88.5%	0.713	1.900
14	2.38	2.39	84.3%	1.270	1.460
15	2.59	3.17	81.3%	1.960	1.880
16	2.89	3.25	84.8%	1.217	6.920
17	3.33	3.41	80.6%	1.170	8.377
18	3.62	3.71	79.8%	1.700	5.300
19	3.90	3.74	76.5%	2.683	8.582

Δ takes positive values throughout Table 12 for most choices of *PI* thresholds across all three phases of an innings. Moreover teams having 100% win rates when $PI \in [0, 0.5)$, irrespective of the phase of an innings, is in terms with the fact that if the Pressure Curve during a run chase reaches values less than 0.5, then the chasing team is highly likely to win, according to Saikia et al.[22] (details in Section "Strategy for controlled run chases using the Markov model"). A negative Δ for $PI \in [3.5, 4)$ during Middle Overs is due to rare occurances of outliers (e.g. Royal Challengers Bangalore vs Chennai Super Kings at M Chinnaswamy Stadium on April 25, 2018), supported on lower sample sizes for matches with extremely high *PI* values. Another notable discrepancy from Table 12 is the rise of A from 0 to 0.6 and 0.9 for $PI \in [4, 4.5)$ and $[4.5, 5)$ respectively, which has been also caused due to a rare outlier: Royal Challengers Bangalore vs Kolkata Knight Riders at M Chinnaswamy Stadium on April 05, 2019.

Table 21. Calibration of predicted probabilities for $PI \in [0, 0.5)$.

Probability bins	Predicted probability	Observed frequency	Calibration error
0.0–0.1	0.0438	0.0498	0.006
0.1–0.2	0.159	0.145	0.014
0.2–0.3	0.280	0.221	0.059
0.3–0.4	0.359	0.374	0.015
0.4–0.5	0.421	0.436	0.015
0.5–0.6	0.588	0.537	0.051
0.6–0.7	0.609	0.673	0.064
0.7–0.8	0.747	0.766	0.019
0.8–0.9	0.852	0.886	0.034
0.9–1.0	0.915	0.982	0.067

Table 22. Calibration of predicted probabilities for $PI \in [0.5, 3.5)$.

Probability bins	Predicted probability	Observed frequency	Calibration error
0.0–0.1	0.0451	0.0448	0.0003
0.1–0.2	0.167	0.152	0.015
0.2–0.3	0.272	0.254	0.018
0.3–0.4	0.354	0.340	0.014
0.4–0.5	0.487	0.448	0.039
0.5–0.6	0.575	0.604	0.029
0.6–0.7	0.622	0.661	0.039
0.7–0.8	0.749	0.783	0.034
0.8–0.9	0.855	0.871	0.015
0.9–1.0	0.914	0.972	0.058

Table 23. Brier Score and ECE for calibration of probabilities in Tables 21 and 22.

Pressure Index	Brier Score	Expected Calibration Error
$PI \in [0, 0.5)$	0.0184	0.00072
$PI \in [0.5, 3.5)$	0.0245	0.00031

Besides the abundance of positive Δ values throughout Table 12, the differences (Δ) are significantly large for higher *PI* values as the innings progresses, i.e. towards the Death Overs. A deeper partition of the innings, based on Overs 16 to 19 (see Table 13) clearly shows teams playing in home conditions have historically shown an edge in handling pressure, towards the final stages of run chases, more than those playing in away conditions. Note that the choice of Overs 16 to 19 instead of taking the Pressure Index at the end of Over 20 into account, is based on the fact that *PI* values at the end of Over 20 is largely concentrated at 0 for successfully completed run chases and around too high values for unsuccessful run chases, leading to an extremely heavy-tailed distribution which has no useful interpretation for predictive purposes.

Tournament-specific model performance. Owing to the variability of playing conditions, competitive intensity and other deciding factors, it could be of interest to check how the model behaves in terms of predictive performance for different tournaments.

Denote *Coverage percentage* to be the percentage of times the actual Pressure Index lies within the 95% confidence interval of the predicted Pressure Index and *Markov Usage percentage* to be the percentage of predicted *PI* values that use the Markov transition matrix (actual observed data) over the Gamma distribution fallback (statistical approximation). Table 14 shows the predictive performance of a few popular tournaments based on the Markov model (see details in Table 24 for the entire list).

From Table 14, one can note a significant difference in MAE and RMSE between leagues played in Asia (IPL, PSL, Syed Mushtaq Ali Trophy etc. has lower MAE and RMSE) and those played outside Asia (T20 Blast, CPL, SA20 League etc. has higher MAE and RMSE). This discrepancy might occur due to various external factors like kind of pitches, dimensions of grounds etc. leading to differences in scoring rates between tournaments and so, it may be of interest to construct Markov models based on datasets of a specific tournament (equivalently, league or competition) and check how they perform in terms of predictive responses for an independent dataset, i.e. a different tournament.

Tables 15 and 16 clearly show several examples of how predictive performances can vary depending on contextual factors involved in a tournament.

Applying the Markov model on Pressure Index

Based on our findings so far, we can predict the Pressure Index values with 0.1 precision, along with suitable confidence intervals supported by phase-wise Gamma distribution fits, at the end of every over during the second innings of a Twenty20 match. Consider the Twenty20 match between Pakistan and West Indies at National Stadium, Karachi on April 03, 2018. Pakistan won the match by 8 wickets while chasing a target of 154 runs and our model captured a fair amount of the actual *PI* values concentrated around the corresponding predicted *PI* values.

From Table 17, it can be observed that 12 out of the 14 predictions have been successful, in the sense that the actual *PI* lie within the 95% confidence interval of the corresponding predicted *PI*, for the model with 0.01 precision, which improves to 13 out of 14 successful predictions for the model with 0.1 precision. For some more assessment, we look at a rather closely contested Twenty20 match played between Chennai Super Kings and Delhi Capitals at Dubai International Cricket Stadium on October 10, 2021. Inspite of the Pressure Index never reducing below 1 throughout the second innings, Chennai Super Kings chased down a target of 173 runs in the 20th over and our model performed fairly well throughout.

Teams win only 19.1% of matches while entering the 20th over of a run chase with a Pressure Index higher than 3.5, which is even lower (13.6%) for matches played until 2021. Based on this fact, the run chase in Table 18 is closer to an outlier than to a general representative of the data, inspite of which the model exhibits 13 out of 16 successful predictions on both a 0.01 scale and 0.1 scale of precision, until the start of the last over.

The above examples show how the indices can be monitored throughout the innings and how continuously revising the probabilities of keeping the Pressure Index within a suitable range (see details in Section "Strategy for controlled run chases using the Markov model") helps a team

remain in control during a run chase in Twenty20 cricket. A pseudo-code to calculate *PI* during run chases, from ball by ball data, is given in Algorithm 7.

Strategy for controlled run chases using the Markov model

It has been observed from extensive studies that if the Pressure Curve during a run chase reaches values less than 0.5, then the chasing team is highly likely to win. Tables 12 and 13 have also supported this fact in our dataset. From Saikia et al.,[22] it is understood that

$$\mathbb{P}(\text{match won by team batting second} \mid 0 < PI < 0.5) = 1, \tag{17}$$

and for the different subintervals of *PI* values between [0.5, 3.5], viz. $[0.5, \lambda_{i_1}]$, $[\lambda_{i_1}, \lambda_{i_2}]$, , $[\lambda_{i_m}, 3.5]$, the values of

$$\mathbb{P}(\text{match won by team batting second} \mid \lambda_i < PI < \lambda_j)$$

generally start with 1 but keep on decreasing until they eventually reach 0. An immediate strategy for teams batting second in Twenty20 matches would be to try to bring down the Pressure Index as low as possible, and effectively below 0.5, at some stage during the innings in order to guarantee a win. However, based on our findings in Section "Methodology", it might be of interest to obtain situation-specific thresholds for Pressure Index which enhances the probability of successfully completing a run chase.

The recommendations for a team on which Pressure Index thresholds one should try to approach during a run chase, have been categorized as four zones, viz. Target Zone, Acceptable Zone, Risky Zone and Avoid Zone, in order of diminishing favourability based on historical win rates and predictive behavior of the phase-wise Markov model. The terminology 'Zone' has been used to keep parity with 'Pressure Zone' as defined by Saikia et al.[22]

Note that the 'Target zone' and 'Acceptable zone' of *PI* thresholds tend to enlarge as the innings progresses. The unbounded intervals $[2.5, \infty)$ in Table 19 are practically not unbounded (see Saikia et al.[22] for methods to control abruptly rising Pressure Index towards the end of an innings during unsuccessful run chases) and should not be a matter of interest for our purpose.

Table 20 shows the mean of actual and predicted Pressure Index values fairly match, along with significantly high coverage percentages for every over of a Twenty20 innings for our test dataset. The over-wise MAE and RMSE are observed to be in parity with phase-wise MAE and RMSE shown in Table 11. On the other hand, relatively higher deviations and smaller coverage probabilities towards the end of the innings are expected to be caused due to the concentration around 0 and too high *PI* values, denoting higher uncertainity of run chases during Death Overs.

Definition 4 Brier score is a strictly proper scoring rule[7] that measures the accuracy of probabilistic predictions. For unidimensional predictions, it is equivalent to the mean squared error as applied to predicted probabilities.

$$Brier\ Score = \frac{\sum(\text{Predicted probability} - \text{Actual outcome})^2}{\text{Total number of predictions}} \tag{18}$$

Definition 5 Expected Calibration Error (ECE) is a measure of quantification of the discrepancy between predicted probabilities and empirical outcome frequencies by aggregating calibration error over a partition of the probability space.

$$ECE = \sum_{k=1}^{\text{Number of bins}} \left(\frac{\text{Number of predictions in } k^{\text{th}} \text{ bin}}{\text{Total number of predictions}} \right) |p_k - o_k| \tag{19}$$

where, p_k = Predicted probability in k-th bin, o_k = Observed frequency in k-th bin.

Tables 21 and 22 show fairly good predictive performance for probabilities of Pressure Index lying in [0, 0.5) and [0.5, 3.5), along with very low Brier Score and ECE (see Table 23), which strengthens the motivation to frame a desired strategy from a coach or team's perspective during run chases in Twenty20 cricket.

Conclusion

The work focuses on the use of higher-order Markov chains to analyze the extent to which the current value of pressure, quantified by the Pressure Index (*PI*), influences future situations of run chases in limited overs cricket, primarily Twenty20 cricket. Previously, no study has addressed the extent to which the current match situation influences future situations. This approach demonstrates that a third order Markov chain is capable of capturing the complex, time-dependent patterns of *PI* values, where the current pressure state depends on the immediate previous three values of the Pressure Index.

The current methodology utilises a vast dataset of 6537 Twenty20 matches spanning over a period of 22 years to build robust transition matrices, developing an immediate and real-time actionable tool. Crucially, this model is the first to use Pressure Index transitions not only for prediction of outcomes (viz. WASP by Hogan et al.[17] or Duckworth-Lewis methods), but also paves the way for understanding the strategy of successful run chases by analyzing historical patterns for a large dataset.

The concept of computing the expected value of the Pressure Index for the next over, given the current match situation, allows the coaching staff and players to set immediate mini-targets so that the PI values remain between 0.5 to 1, (below 0.5 may not always be practically achievable), or at times within 1.5 (see Table 19) as early as possible—which the analysis depicts as a reasonable condition for a successful run-chase. This dynamic and real-time outcome of the study, with over-by-over tactical guidelines, is an innovative approach to the study.

Direction of future research

Tracing Pressure curves[3] and revising Pressure Index values with the help of the proposed Markov model helps to build a strategy to keep teams batting second in Twenty20 matches in control throughout the chase. However, several potential developments to the model can be considered as part of future research:

i.Different teams have different patterns of a run chase. For example, teams with greater batting depths might score at a faster rate even after the fall of a wicket and still succeed but the same strategy might not work for teams with lesser batting depths. Therefore, errors are expected to occur when using the transition probabilities corresponding to one team to predict Pressure Index values for a team with a different composition.

ii.Besides the effects of contextual factors like home ground advantage and tournament-specific models discussed in Section "Contextual Factors in Pressure Index Dynamics", external factors like venues, psychological factors like player-player coalition or player-coach coalition etc. might also play key roles in run chases in Twenty20 matches. Modification of the model to incorporate such external factors, using small sample methodologies, might increase the accuracy while predicting Pressure Index values.

iii.The model could be improved by considering the differences in the qualities and strengths of the teams involved in a match. Apart from the external factors, the ability of teams winning close contests, strengths of oppositions etc. might affect a team's performance. For example, a Pressure Index of say, 1.42, at some stage of the innings against a stronger opposition may have a different impact compared to that against a weaker opposition. In matches between a top-ranked team and a lower-ranked team, the former is expected to capitalize towards the end of the match even from less favorable situations, primarily due to the significant difference in experience possessed by the two teams.

iv.The formula in (6) restricts the usage of Pressure Index to second innings only. Certain variance stabilizing transformations (suggested by Saikia et al.[22]), where $\frac{CRRR}{IRRR}$ is raised to a power $(\frac{CRRR}{IRRR})^{\alpha}$ for some suitable choice of α, help to restrict the abrupt rise in PI values at later stages of the innings during unsuccessful run chases. Such transformations can be applied in constructing a similar Markov model if one intends to use it for prediction purposes for teams batting first as well.

v.The methodology used in this study could be extended to other formats of limited overs cricket, for example One-Day Internationals, with suitable modifications.

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Ethical considerations

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Data availability

The datasets obtained, generated and analyzed during the study are available for free in public domain at https://www.espncricinfo.com and https://cricsheet.org/matches/.

Notes

1. The data for the results of these matches have been collected from https://stats.espncricinfo.com/ci/engine/stats/index.html

2. Data for these matches have been collected from https://www.espncricinfo.com and https://cricsheet.org/matches/
3. This count included repeated transitions as well. The number of unique transitions are listed in Table 2.
4. The choice of the date is based on the fact that the International Cricket Council (ICC) implemented major changes to rules in One Day Internationals, viz. elimination of batting powerplay, free hits for all kinds of no-balls etc, which brought a significant change in scoring rates in One Day Internationals post July 2015.
5. The values corresponding to AIC, BIC and Kolmogorv-Smirnov tests for each of the three distributions in the tables have been calculated using `fitdistrplus` and `MASS` packages in R.
6. The density plots corresponding to the three phases of a Twenty20 innings have been constructed using `fitdistrplus` and `ggplot2` packages in R.
7. A scoring rule S is said to be proper, relative to a convex class $\mathcal{F}$ of probability measures, if its expected score is minimized when the forecasted distribution matches the distribution of the observation, i.e. $\mathbb{E}_{Y\sim Q}[S(Q, Y)] \leq \mathbb{E}_{Y\sim Q}[S(F, Y)]$ for all $F, Q \in \mathcal{F}$. It is strictly proper if this equation holds with equality if and only if $F = Q$.

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Appendix A
A.1 Necessary tables

Table 24. Predictive performance of the Markov model for some tournaments/series.

	Competition	MAE	RMSE	Coverage	Markov Usage
Tournaments	T20 Internationals	1.7749	4.0912	82.43%	93.64%
	Abu Dhabi T20 League	7.1149	13.9755	82.35%	94.12%
	Africa Regional T20 Cup	6.2063	16.5576	82.55%	95.22%
	Bangabandhu T20 Cup	3.1105	4.7792	87.96%	95.24%
	Bangladesh Premier League	1.0662	3.5498	80.58%	96.12%
	BBL	1.8664	3.1107	84.43%	95.92%
	CPL	5.5460	8.9030	84.31%	95.10%
	CSA T20 Challenge	3.3286	4.4902	84.16%	95.52%
	Dhaka T20 League	4.8326	6.5975	81.73%	94.31%
	Emerging Teams Asia Cup	1.9141	3.7764	84.05%	93.25%
	ILT20	1.0259	2.7928	83.86%	95.48%
	IPL	0.8885	7.8697	84.16%	95.86%
	Lanka Premier League	3.9791	9.9558	84.62%	95.78%
	Major League Cricket	2.0930	5.1130	83.61%	95.48%
	National T20 Cup	5.5214	8.1287	84.90%	95.41%
	PSL	0.6081	3.6784	84.53%	95.93%
	Ram Slam T20 Challenge	2.1653	5.0489	81.12%	96.76%
	SA20 League	6.3580	13.7701	84.16%	93.70%
	SLC Twenty-20 Tournament	4.3099	9.4595	82.05%	96.46%
	Super Smash	1.5733	2.8123	84.49%	95.06%
	Super T20 Provincial Tournament	5.429	12.769	92.80%	93.07%
	Syed Mushtaq Ali Trophy	1.6174	3.3080	84.97%	95.66%
	T20 Blast	6.4472	14.1590	83.37%	95.90%
Few randomly picked series	Afghanistan A in Oman T20s	7.8297	5.8597	78.79%	96.97%
	Australia v Sri Lanka 2022	0.4093	1.1497	84.29%	92.86%
	BBL 2023	0.7681	2.7041	87.59%	97.19%
	CPL 2023	4.1487	3.0939	84.75%	94.92%
	CPL 2024	3.9750	4.8647	84.25%	95.00%
	ICC T20 WC Warm-up 2022	1.3045	3.0810	85.02%	87.50%
	Ireland A in Bangladesh T20Is	0.2406	0.4900	100.00%	100.00%
	New Zealand v England 2019	1.0183	6.7465	88.24%	88.24%
	SA20 League 2024	5.3580	7.7701	78.16%	93.70%
	SL2020	4.0565	6.1665	79.49%	94.12%
	South Africa A in Zimbabwe T20Is	1.0273	4.6591	83.33%	89.58%
	Syed Mushtaq Ali Trophy 2018–19	0.9208	1.9607	83.78%	99.10%
	West Indies A in Nepal T20s	1.1591	5.0552	84.67%	87.65%

A.2 Pseudo codes for useful algorithms

Algorithm 1 Construction of State Transitions from Pressure Index

```
for each match m in dataset do
    Extract PI values: [PI_1, PI_2, ..., PI_T]
    for t = k + 1 to T do
        previous state ← (PI_{t-k}, ..., PI_{t-1})
        next state ← PI_t
        Record transition: previous state → next state
        Increment count[previous state, next state]
    end for
end for
```

Algorithm 2 Determine optimal Markov chain order

```
Require: Dataset of M matches with PI sequences
Require: Maximum order k_max
Ensure: Optimal order k*
Split data into 80% training and 20% test sets
for k = 1 to k_max do
    Initialize transition count matrix N^(k)
    N_transitions ← Σ_{m=1}^{M} (T_m − k)                    ▷ Equation 12
    for each match m in training set do
        for t = k + 1 to T_m do
            Extract previous state: (s_{t-k}, ..., s_{t-1})
            Extract current state: s_t
            Increment N^(k)[(s_{t-k}, ..., s_{t-1}), s_t]
        end for
    end for
    Calculate m_k = number of unique transitions
    Calculate log-likelihood: L_k ← Σ_{i=1}^{N_transitions} log P(s_t^(i) | previous state^(i))
    Calculate AIC_k ← 2m_k − 2L_k                            ▷ Equation 15
    Calculate BIC_k ← m_k log(N_transitions) − 2L_k          ▷ Equation 16
    Calculate parameter-to-observation ratio: ρ_k ← m_k / N_transitions
end for
Select k* ← arg min_k {AIC_k, BIC_k : ρ_k < 1}
    return k*
```

Algorithm 3 Discretize Pressure Index values

Require: Continuous PI values $\{PI_1, PI_2, \ldots, PI_n\}$
Require: Precision level δ
Ensure: Discretized PI values $\{PI'_1, PI'_2, \ldots, PI'_n\}$

```
for each PI_i in dataset do
    if PI_i < 0 then
        PI'_i ← 0                                   ▷ Censoring negative values
    else
        PI'_i ← δ × ⌊PI_i/δ + 0.5⌋                  ▷ Round to nearest δ
    end if
end for
    return {PI'_1, PI'_2, ..., PI'_n}
```

Algorithm 4 Predict next Pressure Index value

Require: Transition matrix $\mathcal{T}$
Require: Previous states $(s_{t-k}, \ldots, s_{t-1})$
Require: Phase-specific Gamma parameters (α, β)
Require: Confidence level γ
Ensure: Predicted PI value $\hat{s}_t$ with confidence interval $[\hat{s}_t^{lower}, \hat{s}_t^{upper}]$

```
if (s_{t-k}, ..., s_{t-1}) exists in T then
    Extract all possible next states: {s_t^(1), s_t^(2), ..., s_t^(n)}
    Extract corresponding probabilities: {p_1, p_2, ..., p_n}
    Calculate expected value: ŝ_t ← Σ_{i=1}^{n} p_i · s_t^(i)
else
    Calculate sum: S ← Σ_{j=t-k}^{t-1} s_j
    Search for states in T where Σ_{j=t-k}^{t-1} s'_j = S
    if matching states found then
        Calculate expected value from matching transitions
    else
        Use Gamma distribution fallback: ŝ_t ← α/β                ▷ Mean of Gamma
    end if
end if
Calculate confidence interval using Gamma quantiles:
ŝ_t^lower ← Gamma^{-1}((1-γ)/2; α, β)
ŝ_t^upper ← Gamma^{-1}((1+γ)/2; α, β)
    return ŝ_t, [ŝ_t^lower, ŝ_t^upper]
```

Algorithm 5 Fit phase-wise distributions to Pressure Index

Require: PI values for phase $\phi \in \{\text{Powerplay, Middle, Death}\}$
Require: Distribution families $\mathcal{D} = \{\text{Gamma, Exponential, Weibull}\}$
Ensure: Best-fit distribution parameters for phase ϕ

```
for each distribution d ∈ D do
    Estimate parameters θ_d using Maximum Likelihood Estimation
    Calculate log-likelihood: L_d
    Calculate AIC_d ← 2|θ_d| − 2L_d
    Calculate BIC_d ← |θ_d| log(n) − 2L_d
    Perform Kolmogorov-Smirnov test: (D_d, p_d) ← KS-test(PI, d, θ_d)
end for
Select d* ← arg min_d{AIC_d} subject to p_{d*} > 0.05
    return Distribution d* with parameters θ_{d*}
```

Algorithm 6 Parametric Bootstrap for Kolmogorov-Smirnov Goodness-of-Fit

Require: Observed data $x_1, \ldots, x_n$; distribution family $\mathcal{D}$ (e.g., Gamma)
Ensure: K–S statistic D_{obs} and bootstrap p-value $\hat{p}$

```
Fit D to observed data to obtain parameter estimate θ̂
Compute D_obs = sup_x |F_n(x) − F(x; θ̂)|
Choose bootstrap size B
for b = 1 to B do
    Generate sample x_1^(b), ..., x_n^(b) ~ D(θ̂)
    Fit D to bootstrap sample to obtain θ̂^(b)
    Compute D^(b) = sup_x |F_n^(b)(x) − F(x; θ̂^(b))|
end for
Compute bootstrap p-value: p̂ = (1/B) Σ_{b=1}^{B} 1{D^(b) ≥ D_obs}
    return D_obs, p̂
```

Algorithm 7 Calculate Pressure Index at any stage during a run chase

Require: Target T
Require: Total balls B
Require: Runs scored R'
Require: Balls faced B'
Require: Wickets fallen $w_1, w_2, \ldots, w_k$
Require: Resources used RU
Ensure: Pressure Index PI

```
Calculate Initial Required Run Rate: IRRR ← (T×6)/B
Calculate Current Required Run Rate: CRRR ← ((T−R')×6)/(B−B')
Initialize wicket weight sum: Σ w_i ← 0
for each fallen wicket at position i do
    Σ w_i ← Σ w_i + w_i                                  ▷ Use weights from Table 1
end for
Calculate: PI ← (CRRR/IRRR) × (1/2) [e^(RU/100) + e^(Σ w_i / 11)]
    return PI
```

Algorithm 8 Generate over-by-over strategic recommendations

Require: Current over t
Require: PI history $(PI_1, \ldots, PI_t)$
Require: Target T
Require: Wickets lost w
Require: Home/Away status
Require: Trained phase-wise models $\{\mathcal{M}_{\text{PP}}, \mathcal{M}_{\text{MO}}, \mathcal{M}_{\text{DO}}\}$
Ensure: Strategic recommendation and target PI range

```
Determine current phase φ based on over t
Select appropriate model M_φ and thresholds from Table 19
Predict next PI: P̂I_{t+1} ← M_φ(PI_{t−2}, PI_{t−1}, PI_t)
if P̂I_{t+1} ∈ [0, 0.5) then
    Recommendation ← "Target Zone - Maintain aggressive scoring"
else if P̂I_{t+1} ∈ [0.5, 1.5) then
    Recommendation ← "Acceptable Zone - Continue current approach"
else if P̂I_{t+1} ∈ [1.5, 2.5) AND phase = Death Overs then
    Recommendation ← "Acceptable/Risky - Accelerate carefully"
else
    Recommendation ← "Avoid Zone - High risk, need immediate acceleration"
    Calculate required run rate adjustment
end if
Display: Current PI, Predicted PI range, Recommendation, Required runs per over
    return Recommendation with actionable targets
```